Analyze Graphs of Relationships Between Two Variables

Read the rate of change from a graph and use it to predict a value you cannot see.

Name _____ **Date** _____

Tonight's spark

A school printer graph has the points (1, 13), (2, 26) and (3, 39). The x-values are minutes and the y-values are pages. The graph stops at 3 minutes. How many pages print in 10 minutes?

Guess first: is it more than 100 pages, or less?

Minutes (x)	Pages (y)
1	13
2	26
3	39
10	?

13 each

Our class model: read the rate, then predict

Try it

Work it out on paper. Write one answer on each line.

1 **A graph has the points (1, 4), (2, 8), (3, 12). What is the rate of change?**

A. 1 **B.** 4 **C.** 8 **D.** 12

My answer _____

2 **The same graph has the points (1, 4), (2, 8), (3, 12). What is y when x = 4?**

A. 16 **B.** 13 **C.** 20 **D.** 12

My answer _____

3 **A graph has the points (1, 3), (2, 6), (3, 9). What is y when x = 10?**

A. 13 **B.** 39 **C.** 10 **D.** 30

My answer _____

4 **The point (5, 15) is on a graph. What does the 5 stand for?**

A. the output **B.** the origin **C.** the independent value (input) **D.** the rate of change

My answer _____

5 **A graph passes through (0, 0) and (2, 10). What is the unit rate, or y for each 1 of x?**

A. 2 **B.** 10 **C.** 12 **D.** 5

My answer _____

Talk about it

What is the rate of change on your graph? How did it help you find a point that is not drawn?

Home mission

Pretend a printer makes 13 pages each minute. Ask your student to say the point for 5 minutes, with minutes first. Then ask how they found it.

If we get stuck, ask

- What does this point mean in real words?
- How much does y climb each time x goes up by one?
- Which number in the pair did you write first?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/9-2-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it